

# All-Angle Fixed Sets for the Planar Harmonic Strobe

Route-A structural certificate C178

## Abstract

For the planar oscillator  $H = (q^2 + p^2)/2$ , the time- $\theta$  strobe is a rotation. Its  $n$ th iterate fixes only the origin unless  $n\theta \in 2\pi\mathbb{Z}$ , when it fixes the whole plane. Thus irrational  $\theta/(2\pi)$  gives the Artin–Mazur zeta  $(1 - z)^{-1}$ , whereas every rational angle has an uncountable resonant fixed set and no classical Artin–Mazur series. This exact dichotomy rejects a uniform periodic-orbit determinant interpretation; no arithmetic or target-spectrum claim is made.

**Keywords:** harmonic oscillator; stroboscopic rotation; fixed sets; resonance; Artin–Mazur zeta; Route A.

## 1 Frozen flow and fixed sets

Hamilton’s equations give

$$T_\theta(q, p) = (q \cos \theta + p \sin \theta, -q \sin \theta + p \cos \theta), \quad T_\theta^n = T_{n\theta}.$$

A nonidentity rotation fixes only  $(0, 0)$ , while the identity fixes  $\mathbb{R}^2$ . Therefore

$$\text{Fix}(T_\theta^n) = \begin{cases} \mathbb{R}^2, & n\theta \in 2\pi\mathbb{Z}, \\ \{(0, 0)\}, & n\theta \notin 2\pi\mathbb{Z}. \end{cases}$$

If  $\theta/(2\pi)$  is irrational, every fixed count is one and

$$\zeta_{\text{AM}}(z) = \exp\left(\sum_{n \geq 1} \frac{z^n}{n}\right) = \frac{1}{1 - z}.$$

If  $\theta/(2\pi) = a/b$  is reduced, the whole plane is fixed exactly when  $b \mid n$ . Its uncountable cardinality is not a finite Artin–Mazur coefficient, so the classical series is undefined. This includes the zero-angle edge  $b = 1$ .

The reflection  $S(q, p) = (q, -p)$  is involutive and satisfies  $ST_\theta S = T_\theta^{-1}$ .

## 2 Decision boundary

The finite ledger supplied with the paper is a regression sentinel for the rotation formula, not a proof. The result uses no target zero or prime table, fitted parameter, arithmetic local datum, Euler factor, root number, automorphy object, or Hilbert–Pólya assertion. The scope is NO\_BAD\_EULER\_OR\_ROOT\_NUMBER, and Route B is unauthorized.

**Declarations.** All data and code are included. No external dataset, private data, or human or animal subject is involved. The anonymous contribution covers conceptualization, proof, software, validation, and writing. No funding or competing interest is declared. AI assistance was used for drafting and code generation; claim-bearing formulas were separately reconstructed by distinct internal code paths, and no external reviewer or acceptance score is represented.